

Math 22 Curriculum

Geometry | Middlesex School Mathematics | Resource: Math 22 Geometry Packet

COURSE AT A GLANCE

Review Content

REVIEW

No mapped OpenStax section

UNIT LEARNING OBJECTIVES

- I can solve a linear equation arising from a geometric relationship.
- I can solve a proportion and interpret the resulting scale factor.
- I can simplify an exact square root and approximate it when needed.
- I can calculate distance, midpoint, and slope on the coordinate plane.
- I can substitute measurements into a formula with consistent units.

Geometry Basics

REQUIRED

1.1 Points, Lines, Planes; 1.2 Segments, Measurement, Addition, Midpoints, Bisectors; 1.3 Angles, Measurement, Addition, Bisecting, Pairs; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular; 1.5 Parallel-Line Converses and Proofs

UNIT LEARNING OBJECTIVES

- I can name and represent points, lines, planes, segments, and rays using correct notation.
- I can identify collinear, coplanar, intersecting, and concurrent figures.
- I can calculate segment lengths using betweenness and the segment-addition postulate.
- I can determine a midpoint or use a segment bisector relationship.
- I can classify and measure angles.
- I can calculate angle measures using angle addition and angle bisectors.
- I can identify and use complementary, supplementary, vertical, and linear-pair relationships.
- I can distinguish a definition, postulate, theorem, converse, and counterexample.
- I can determine angle relationships formed by parallel lines and a transversal.
- I can use angle relationships to determine whether lines are parallel.
- I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.
- I can construct a logical geometric argument from stated facts and accepted results.

Triangle Congruence and Proof

REQUIRED

2.1 Triangles, Sides, and Angles; 2.2 Congruent Polygons; SAS; 2.3 SSS and HL; 2.5 ASA and AAS; 2.4 Isosceles and Equilateral Triangles

UNIT LEARNING OBJECTIVES

- I can classify triangles by sides and angles.
- I can use triangle angle-sum and exterior-angle relationships.
- I can write a congruence statement with corresponding vertices in the correct order.
- I can determine whether triangles are congruent by SAS.
- I can determine whether triangles are congruent by SSS.
- I can determine whether right triangles are congruent by HL.
- I can determine whether triangles are congruent by ASA or AAS.
- I can apply isosceles and equilateral triangle theorems and their converses.
- I can use corresponding parts of congruent triangles after proving congruence.
- I can write a structured proof involving triangle congruence.

Quadrilaterals, Perimeter, and Area

REQUIRED

3.1 Polygon Angles, Perimeter, and Area; 3.2 Parallelograms; 3.3 Rhombi, Rectangles, Squares; 3.4 Trapezoids; 3.5 Kites and Composite Polygons

UNIT LEARNING OBJECTIVES

- I can calculate interior or exterior angle measures of a polygon.
- I can calculate perimeter and area of common polygons.
- I can apply properties of parallelograms to determine unknown measures.
- I can determine whether a quadrilateral is a parallelogram.
- I can apply properties of rectangles, rhombi, and squares.

Quadrilaterals, Perimeter, and Area (Continued)

REQUIRED

Similarity and Proportional Reasoning

REQUIRED

4.1 Dilations and Similarity; 4.2 Similar Polygons, Perimeter, and Area; 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS; 4.5 Proportionality Theorems

UNIT LEARNING OBJECTIVES - CONTINUED

- I can classify a quadrilateral from its stated or marked properties.
- I can apply trapezoid and trapezoid-midsegment relationships.
- I can apply properties of kites.
- I can calculate the perimeter or area of a composite polygon.
- I can justify a quadrilateral conclusion using definitions and theorems.

UNIT LEARNING OBJECTIVES

- I can determine image lengths and coordinates under a dilation.
- I can identify the scale factor of a dilation or similarity relationship.
- I can write a similarity statement with corresponding vertices in the correct order.
- I can use proportional corresponding sides to solve similar-polygon problems.
- I can relate similarity scale factor to perimeter and area scale factors.
- I can prove triangles similar by AA, SSS, or SAS.
- I can apply triangle proportionality theorems and their converses.
- I can solve and justify an indirect-measurement or other similarity application.

Right Triangles

REQUIRED

5.1 Pythagorean Theorem and Triangle Inequalities; 5.2 Special Right Triangles; 5.3 HL Similarity and Right-Triangle Similarity; 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

UNIT LEARNING OBJECTIVES

- I can apply the Pythagorean theorem to determine a missing side.
- I can use the converse of the Pythagorean theorem to classify a triangle.
- I can apply triangle-inequality relationships.
- I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.
- I can apply right-triangle similarity relationships.
- I can use geometric-mean relationships created by an altitude to the hypotenuse.
- I can select sine, cosine, or tangent for a right-triangle problem.
- I can determine a missing side or angle of a right triangle.
- I can solve a contextual right-triangle problem.
- I can justify a right-triangle solution using an appropriate theorem or ratio.

Circles

REQUIRED

6.1 Circle Vocabulary, Circumference, Arc Length; 6.2 Circle, Sector, and Segment Area; 6.3 Angle, Arc, and Chord Relationships; 6.4 Tangent, Secant, and Chord Relationships; 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

UNIT LEARNING OBJECTIVES

- I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.
- I can calculate circumference and arc length.
- I can calculate the area of a circle, sector, or segment.
- I can determine angle and arc measures involving central and inscribed angles.
- I can apply relationships among chords, arcs, and distances from the center.
- I can apply tangent-radius and tangent-segment relationships.
- I can determine angle measures formed by chords, secants, or tangents.
- I can determine segment lengths formed by intersecting chords.
- I can determine segment lengths formed by secants or tangents.
- I can solve a multi-step circle problem and justify each relationship used.